

UNIT I

BUILDING MATERIALS

➤ STONES

➤ BRICKS

➤ CEMENT

➤ CONCRETE

➤ STEEL



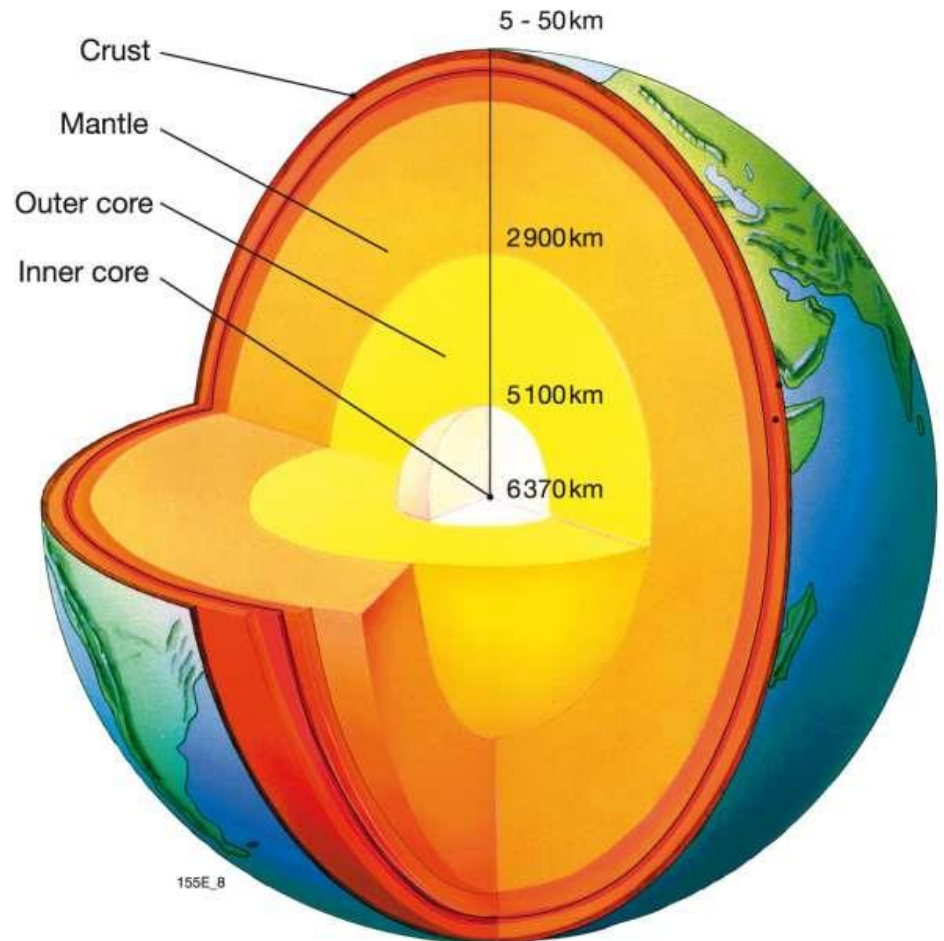
STONES

- Stone is a naturally available material of construction and is obtained from rocks.

Classification of Rocks

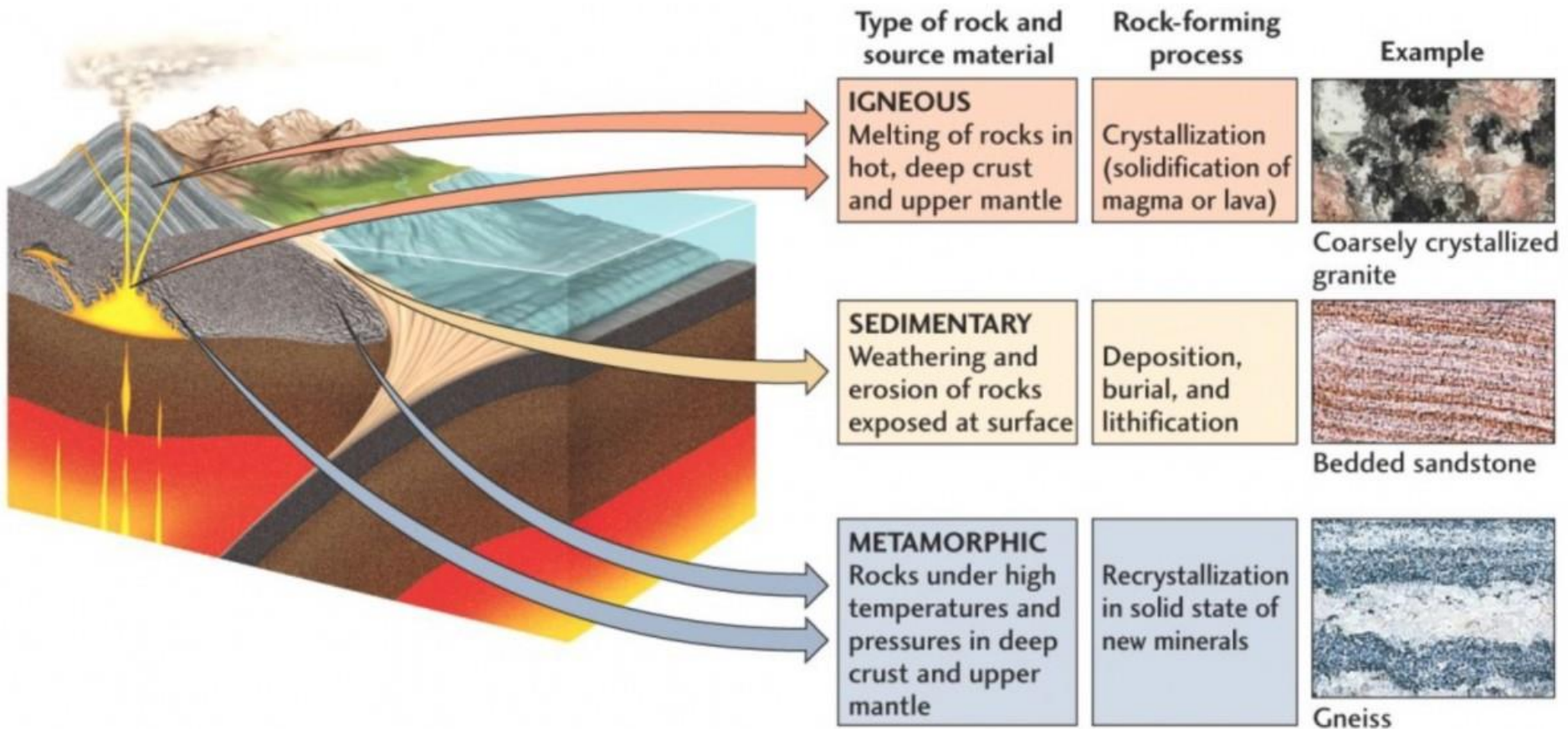
Rocks are classified in following three ways

1. Geological Classification
2. Physical Classification
3. Chemical Classification



Geological classification

- Igneous rocks
- Sedimentary rocks
- Metamorphic rocks



Geological Classification

Based on the mode of formation, the rocks are classified as:

i. Igneous Rocks:

- These rocks are formed by the cooling of molten rocky material called magma which is inside the earth's surface.

eg. Granite.

- If the magma cools at a relatively shallow depth from earth's surface then it is called Hyperbyssal rocks.

eg. Dolerite.

- If the magma cools at earth's surface than it is called volcanic rocks.

eg. Basalt.



Geological Classification contd...

ii. Sedimentary Rocks:

- Sedimentary rocks are formed by the deposition of products of weathering on the pre-existing rock.
 - All the products of weathering are ultimately carried away from their place of origin by the agents of wind, rain, frost, etc.
- eg. Sandstone, Limestone, Gypsum, Gravel etc



Geological Classification contd...

iii. Metamorphic Rocks: When the pre-existing rocks (i.e. Igneous and Sedimentary rocks) are subject to great heat and pressure, they are changed in character and forms metamorphic rocks.

eg. Slate, Marble, Gneisses



Slate



Marble



Gneiss

Physical Classification

- **Stratified Rocks:** These rocks possess planes of stratification and such rocks can easily be split up along these planes.

eg. Sedimentary Rocks.

- **Unstratified Rocks:** These rocks do not exhibit any definite layers or strata.

eg. Igneous rocks.

- **Foliated Rocks:** These rocks have a tendency to be split up in a definite direction only.

eg. Metamorphic Rocks.



Chemical Classification

- **Silicious Rocks:** In these rocks, Silica is the main constituent.
eg. Granite Quartzites etc
- .
- **Argillaceous Rocks:** In these rocks, clay or argil is the main constituent.
eg. Slates, Laterites etc.
- **iii) Calcareous Rocks:** In these rocks, Calcium Carbonate is the main constituent
eg. Lime stones, Marbles etc.

AGGREGATE (stones)



Usage of stones



Paving road



Railway track Ballast



Road Construction

Retaining walls



Dams construction



Water barriers

Buildings





The famous stone at Stonehenge on Salisbury Plain, Wiltshire



Thanjavur Brihadeeshwarar Temple

Architectural aspects



Uses of Stones

- Stones are used as basic material for concrete, moorum of roads, calcareous cements etc.
- Stones are adopted to form paving of roads and foot paths.
- Stones are converted to form basic materials for concrete, base course of roads, artificial stones, hollow blocks, etc.
- Stones are also used as ballast for railway track.
- Stone blocks are used in the construction of bridges, abutments retaining wall, dams etc.

Tests for Stones

The following are the tests conducted on stones to decide construction work

1. Impact test
2. Crushing strength test
3. Attrition test
4. Hardness Test
5. Water absorption test
6. Freeze - thaw test
7. Microscopic test
8. Smith's test

<i>Property</i>	<i>Test</i>	<i>Significance</i>
Gradation	Sieve Analysis (washed for fine aggregates)	Strength in base courses and asphalt mixes, economy in concrete.
Fines Content	Washed Sieve Analysis (fine aggregate) Washing Test (coarse aggregate) Atterberg Limits	Strength in base courses and asphalt mixes, drainage and frost problems in highway bases, economy in concrete.
Relative Density (Specific Gravity) and Absorption	Relative Density	Design calculations, mix adjustments.
Hardness	Los Angeles Abrasion, Deval Abrasion	Wear of surface particles, particle breakage.
Durability	Soundness	Durability, resistance to weathering, freezing.
Particle Shape	Amount of Thin or Elongated Particles	Durability of particle.
Particle Surface	Amount of Crushed Particles	Strength in base courses and asphalt mixes.
Deleterious Particles or Substances	Petrographic Test, Sand Equivalent Test	Durability of particles.
Chemical Stability	Reactivity (concrete aggregates), Stripping (asphalt aggregates)	Durability of concrete and asphalt.

Aggregate impact test

IS : 2386 (part 4)

- Sieve the aggregate pass through 12.5 mm and retain in 10 mm sieves (A)
- A metal hammer weighing 13.5 to 14.0 kg the lower end being cylindrical in shape, 50 mm long, 100.0 mm in diameter is allowed to fall from 380 mm height for 15 times.
- The crushed aggregate is removed from the cup and sieved on 2.36 mm. IS sieve (B)
- Aggregate impact value = $(B/A) \times 100\%$



<20%	Exceptionally Strong
10 – 20%	Strong
20-30%	Satisfactory for road surfacing
>35%	Weak for road surfacing

Aggregate crushing value

IS: 2386 (Part IV) – 1963.

- A specimen of weight 3 kg (A) filled in cylinder and tested in a compression testing machine.
- The rate of loading is 4.0 N / mm² per minute.
- The maximum load of 40 tonnes for minute is maintained and the load is released.
- Sample is then sieved through a 2.36 mm IS Sieve (B)
- Aggregate crushing value
= $(B/A) \times 100\%$.

Not more than 25 % for wearing surfaces

Not more than 45 % for non-wearing surfaces



Attrition test:

- In this test, some known weight of stone pieces are taken and put in the Deval's attrition test cylinder.
- The cylinder is rotated about its horizontal axis for 5 hrs at the rate of 30 RPM. Then the contents in the cylinder are sieved by 1.5 mm sieve.

$$\text{Percentage of Wear} = \frac{\text{Initial Weight}}{\text{Loss in Weight}} \times 100$$



Deval attrition testing cylinder

Los Angeles Abrasion test

- 5 kg of aggregate sieved as per the norms given in IS : 2386 (Part 5)
- Steel balls 11 or more will be dropped and rotated at 30 rpm for 1000 revolutions.
- The percentage change in the weight is measured.



Sl. No.	Type of Pavement	Max. permissible abrasion value in %
1	Water bound macadam sub base course	60
2	WBM base course with bituminous surfacing	50
3	Bituminous bound macadam	50
4	WBM surfacing course	40
5	Bituminous penetration macadam	40
6	Bituminous surface dressing, cement concrete surface course	35
7	Bituminous concrete surface course	30



Water absorption test:

- A stone of known weight is immersed in water for 24 hrs.
- Then it is weighed again after 24 hours
- Percentage absorption of water should not exceed 0.6



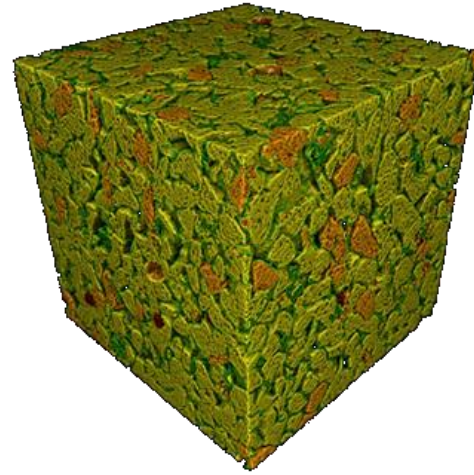
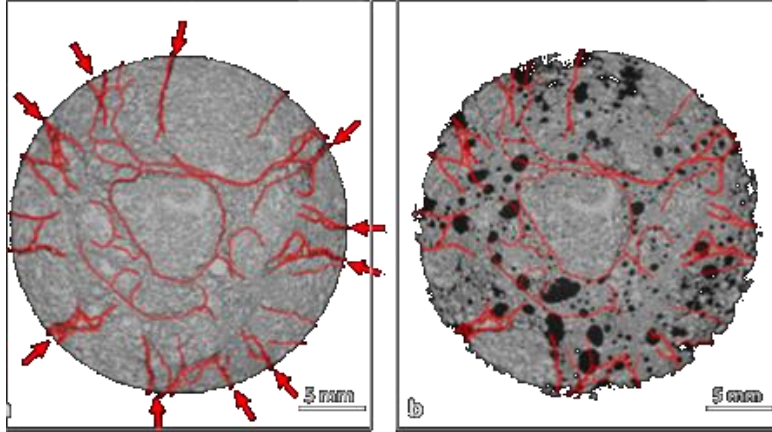
Freezing and thawing test:

- Stone is placed in freezing mixture at 12°C for 24 hours and it is then warmed at atmospheric temperature.
- The procedure is repeated for several times and the behavior of stone is noted.



Microscopic test:

- In this test, thin sections of stone are taken and they are examined in a microscope to study various properties like grain size, mineral constituents etc.



Smith's test:

- Clear water is taken and pieces of stones are placed in it.
- Water is vigorously stirred.
- If the water becomes dirty it indicates the stone contains earthy matter / loose particles.



Qualities of Good Building Stone

A good building stone should have the following qualities:

- Stones must be decent in appearance and be of uniform colour.
- Stones must be durable
- Must be acid resistant and free from any soluble matter.
- Fracture should be sharp and clear.
- It must have a wear less than 3 %.
- It must have a specific gravity greater than 2.7.
- It must have a compact, fine, crystalline structure, and strong

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